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X. Wu, E. Stavitski

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Contrasting Capability of Single Atom Palladium for Thermocatalytic versus Electrocatalytic Nitrate Reduction Reaction

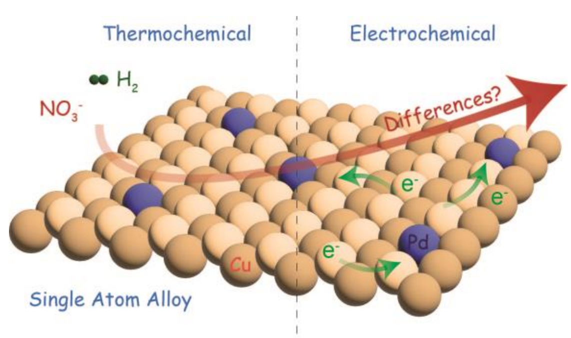
Xuanhao Wu,[1,2‡] Mohammadreza Nazemi,[2‡] Srishti Gupta,[3] Adam Chismar,[3] Kiheon Hong,[4] Hunter P. Jacobs,[4] Wenqing Zhang,[4] Kali Rigby,[2] Tayler Hedtke,[2] Qingxiao Wang,[5] Eli Stavitski,[6] Michael S. Wong,[4] Christopher Muhich,[3,*] and Jae-Hong Kim[2,*]

[1] Department of Environmental Engineering, Zhejiang University, Hangzhou, 310058, China
[2] Department of Chemical and Environmental Engineering, Yale University, New Haven, CT 06511, USA
[3] School for the Engineering of Matter, Transport, and Energy, Arizona State University, Tempe, AZ 85287, USA
[4] Department of Chemical and Biomolecular Engineering, Rice University, Houston, TX 77005, USA
[5] Imaging and Characterization Core Lab, King Abdullah University of Science and Technology, Thuwal, 23955-6900, Saudi Arabia
[6] National Synchrotron Light Source II, Brookhaven National Laboratory, Upton, NY 11973, USA

Corresponding Authors
Jae-Hong Kim: Phone: +1-203-432-4386; fax: +1-203-432-4387; E-mail: jaehong.kim@yale.edu
Christopher Muhich: Phone: +1-480-965-2673; fax: +1-480-727-9321; Email: cmuhich@asu.edu

Abstract. The occurrence of high concentration of nitrate in various water resources is a significant environmental and human health threat, demanding effective removal technologies. Single atom alloys (SAAs) have emerged as a promising bimetallic material architecture in various thermocatalytic and electrocatalytic schemes including nitrate reduction reaction (NRR). This study suggests that there exists a stark contrast between thermocatalytic (T-NRR) and electrocatalytic (E-NRR) pathways that resulted in dramatic differences in SAA performances. Among Pd/Cu nanoalloys with varying Pd:Cu ratios from 1:100 to 100:1, $Pd/Cu_{(1:100)}$ SAA exhibited the greatest activity ($TOF_{Pd} = 2\ min^{-1}$) and highest N_2 selectivity (94%) for E-NRR, while the same SAA performed poorly for T-NRR as compared to other nanoalloy counterparts. DFT calculations demonstrate that the improved performance and N_2 selectivity of $Pd/Cu_{(1:100)}$ in E-NRR compared to T-NRR originates from the higher stability of NO_3^* in electrocatalysis, and a lower N_2 formation barrier than NH due to localized pH effects and the ability to extract protons from water. This study establishes the performance and mechanistic differences of SAA and nanoalloys for T-NRR versus E-NRR.

Keywords: Single atom alloy • Nitrate reduction • Thermocatalysis • Electrocatalysis • Palladium • Copper



Introduction

Nitrate (NO_3^-) is one of the most widespread pollutants in surface and ground water,[1] originating from various sources such as nitrogen-rich fertilizers, domestic and industrial wastewater, landfill leachate, and atmospheric deposition.[2] High concentrations of nitrate in drinking water can lead to adverse health effects including methemoglobinemia (blue baby disorder)[3] and elevated risk of cancer such as non-Hodgkin's lymphoma through the metabolic production of N-nitroso compounds in the body.[4] Together with phosphorus, excess nitrate in surface water also accelerates eutrophication, affecting the ecological balance of aquatic species.[5] For these reasons, the U.S. Environmental Protection Agency (US EPA) has set a maximum contaminant level (MCL) of 10 mg/L for nitrate in drinking water.[6]

Recently, catalytic treatment that enables the conversion of nitrate to benign nitrogen gas (N_2) for environmental remediation or to ammonia (NH_3) for resource recovery has received widespread attention.[7] Two catalytic approaches have been demonstrated: thermocatalytic and electrocatalytic nitrate reduction reactions (T-NRR and E-NRR, respectively). T-NRR, in which hydrogen gas (H_2) acts as the reducing agent, is often catalyzed by bimetallic catalysts consisting of a promoter metal (M_P – *e.g.*, In, Cu, Sn, Zn, Ni) and a noble metal (M_N – *e.g.*, Pd, Pt, Au, Ru), which together contribute to nitrate reduction in multi-step redox processes.[8] Due to their high reactivities, palladium based bimetallic catalysts, such as Pd-In, Pd-Cu, and Pd-Sn, have been mostly investigated since the first introduction by Vorlop and coworkers in 1989.[9] The N_2 and ammonia pathways are shown in Eq. 1 & 2 below, respectively.

$$2NO_3^- + 5H_2 \rightarrow N_2 + 4H_2O \ + 2OH^- \qquad \text{(Eq. 1)}$$

$$NO_3^- + 4H_2 \rightarrow NH_3 + 2H_2O \ + OH^- \qquad \text{(Eq. 2)}$$

In E-NRR, nitrate is reduced electrochemically at the cathode with the electrolyte providing the source of H in the form of reducible protons (Eq. 3 & 4). Monometallic (*e.g.*, Cu, Ag, Au, Ru, Rh, Ir, Pd, Pt, Sn)[10] and bimetallic catalysts (*e.g.*, Cu-Sn,[11] Cu-Pd,[12] Cu-Pt,[13] Cu-Zn,[14] Cu-Ni[15], Rh-Pt[16]) have been reported to be capable of catalyzing E-NRR. Copper has been most applied due to its high E-NRR activity [17] and lower price[18] compared to most other transition or post-transition metals. Under E-NRR conditions, several pathways exist depending on the pH of the environment; the reactions to N_2 or NH_3 for acidic (Eq. 3 & 4) and basic (Eq. 5 & 6) environments are shown below along with their half-cell potentials.[19]

$$NO_3^- + 6H^+ + 5e^- \rightarrow \frac{1}{2}N_2 + 3H_2O \qquad E^0 = 1.25\ V\ vs\ RHE \qquad \text{(Eq. 3)}$$

$$NO_3^- + 10H^+ + 8e^- \rightarrow NH_4{}^+ + 3H_2O \qquad E^0 = 0.88\ V\ vs\ RHE \qquad \text{(Eq. 4)}$$

$$NO_3^- + 3H_2O + 5e^- \rightarrow \frac{1}{2}N_2 + 6OH^- \qquad E^0 = 0.25\ V\ vs\ RHE \qquad \text{(Eq. 5)}$$

$$NO_3^- + 7H_2O + 8e^- \rightarrow NH_3 \cdot H_2O + 9OH^- \quad E^0 = -0.12\ V\ vs\ RHE \quad \text{(Eq. 6)}$$

Among various catalyst combinations, bimetallic Pd/Cu nanoparticle-alloyed catalysts (denoted as $Pd/Cu_{(x:y)}$ nanoalloy) have been shown to effectively catalyze nitrate reduction both thermocatalytically and electrocatalytically.[12, 20, 21] Recently, single atom alloys (SAAs) consisting of an atomically dispersed metal alloyed onto the surface of a host metal have emerged as an alternative catalyst architecture in chemical, energy, and environmental applications due to their high atomic efficiency.[22] We note that single Pd atoms alloyed on Cu nanoparticles (denoted as $Pd/Cu_{(1:y)}$ SAA in this work) have been used for thermocatalytic hydrogenation processes (*e.g.*, styrene and acetylene hydrogenation),[23] NO reduction using CO,[24] and electrochemical reduction of CO_2 and N_2.[25] We also note that the use of single atom Pd instead of Pd nanoparticles would greatly reduce the cost of catalysts, making SAA an appealing material choice for cost-sensitive applications such as water treatment. Past studies suggest that atomically dispersed Pd atoms can effectively dissociate H_2 into H_{ads} which spills onto Cu sites in thermocatalysis.[23] Consequently, we hypothesize that decreasing Pd to SAA in Pd/Cu alloy increases T-NRR performance. Various bimetallic catalysts such as $Pd/Cu_{(x:y)}$,[12, 20] Rh/Pt(100),[26] Cu/Pt(100),[26] Ag/Au(111),[27] Sn/Pt(111),[28] and In/Pt(111)[28] have been also employed to achieve E-NRR, some of which were in SAA form; in Ru/Cu(111), Ru_1 provides active NRR sites and Cu NP sites suppress the hydrogen evolution reaction (HER).[29] However, previous SAAs for NRR including Ru/Cu(111) were NH_3 selective, whereas a N_2 selective catalyst would be preferred for water remediation scenarios. Given the proclivity of Pd nanoparticles to form N_2,[30, 31] we hypothesize that the construction of Pd SAA catalyst is likely to enhance N_2 selectivity. However, the Pd SAA activity has not yet been experimentally demonstrated under either T-NRR or E-NRR. In both T-NRR and E-NRR, the available H sources, pH, and electrons may alter the NRR pathways, activity, and selectivity of $Pd/Cu_{(1:y)}$ SAA.

In this study, we synthesized $Pd/Cu_{(x:y)}$ nanoalloys and $Pd/Cu_{(1:y)}$ SAA, with varying Pd:Cu ratios from 100:1 to 1:100. The performance of different Pd/Cu combinations was evaluated for both T-NRR and E-NRR. We found that the NRR performance differed for the same Pd/Cu catalysts under T-NRR versus E-NRR conditions: *i.e.*, in T-NRR, $Pd/Cu_{(x:y)}$ nanoalloys outperformed $Pd/Cu_{(1:y)}$ SAA, while in E-NRR, $Pd/Cu_{(1:y)}$ SAA outperformed $Pd/Cu_{(x:y)}$ nanoalloys. The $Pd/Cu_{(1:100)}$ SAA's activity increased significantly when switching from T-NRR to E-NRR, with high selectivity towards N_2. Density-functional theory (DFT) calculations were used to build a fundamental understanding of the differences in the $Pd/Cu_{(1:100)}$ SAA performance in T-NRR and E-NRR.

Results and Discussion

Structural characterization of $Pd/Cu_{(x:y)}$ catalysts. A series of $Pd/Cu_{(x:y)}$ catalysts with varying Pd:Cu molar ratios (x:y = 100:1, 20:1, 10:1, 5:1, 3:1, 1:1, 1:3, 1:5, 1:10, 1:20, 1:100) were prepared by the incipient wetness co-impregnation (IWI) method (**Figure S1, Table S1**), which has been previously used to synthesize bimetallic nanoalloys[32, 33] and SAAs.[34] TEM images confirmed the formation of nanoparticles on the mesoporous silica surface after impregnation with Pd and Cu (**Figure S2a**). The average nanoparticle sizes (diameters, nm) obtained via statistical analysis of TEM images showed a U-shape curve when plotted against the Pd/Cu$_{(x:y)}$ ratio, ranging from 100:1 to 1:100 (**Figure S2b**). XRD characterization of Pd/Cu$_{(100:1)}$ indicated a Pd(111) peak at 2θ = 40.1º and a Pd(200) peak at 2θ = 46.8º (**Figure S3**), suggesting high purity of Pd content at this molar ratio.[12] From Pd/Cu$_{(100:1)}$ to Pd/Cu$_{(1:100)}$, a continuous shift of Pd(111) peak to Cu(111) was observed (**Figure S3**),[12, 35] with Pd/Cu$_{(1:10)}$ and Pd/Cu$_{(1:20)}$ showing a Pd/Cu mixture phase at 2θ = 41.1º.[36] This peak-shifting with lower Pd content is likely associated with the decrease in lattice distance (*d*) due to the continuous replacement of Pd by smaller Cu atoms.[12] When Cu content was further increased to Pd/Cu$_{(1:100)}$, distinct Cu nanoparticle phases including Cu(111) at 2θ = 43.3º and Cu(200) at 2θ = 50.4º were observed.[35]

CO-DRIFTS of Pd/Cu$_{(x:y)}$ with increasing Cu wt% confirmed relative Pd and Cu surface concentrations on the catalysts based on changes in the carbonyl stretching (*ν*(CO)) frequencies of adsorbed CO (**Figure 1b**). The spectra of Pd/Cu$_{(100:1)}$ and Pd/Cu$_{(3:1)}$ showed linear CO−Pd interactions at 2090 and 2060 cm^{-1}, and bridging CO−Pd interactions (bridging mode on two contiguous surface Pd atoms) at 1990, 1960, and 1925 cm^{-1} (**Figure S4**).[37, 38] The linear CO peak at 2060 cm^{-1} causes an asymmetric peak shape in the linear CO adsorption region, indicating either Pd corner sites or Pd sites isolated by Cu.[37] The presence of multiple bridging CO peaks indicates CO adsorption on different Pd facets, *i.e.*, Pd(111), Pd(200), and edge sites.[37, 38] A small band associated with CO bound on 3-fold Pd hollow sites (1860 cm^{-1}) was also observed for Pd/Cu$_{(100:1)}$. From Pd/Cu$_{(100:1)}$ to Pd/Cu$_{(3:1)}$, a relatively higher percent of linear interaction versus bridging interaction was observed (**Figure S5**), suggesting that the increased Cu-content caused Pd sites to be more isolated. For Pd/Cu$_{(1:100)}$, a single linear CO−Cu interaction peak was observed at 2125 cm^{-1},[39] with no clear CO−Pd interaction peaks, confirming the absence of large Pd nanoclusters on its surface. Minor CO−Cu interactions were also observed on Pd/Cu$_{(100:1)}$ and Pd/Cu$_{(3:1)}$, indicating Cu is also present on the surface.

To further verify the structural alterations with increasing Cu wt% from Pd/Cu$_{(100:1)}$ to Pd/Cu$_{(1:100)}$, we performed XAS analysis to evaluate the local coordination environment. The Pd K-edge XANES of all the Pd/Cu$_{(x:y)}$ catalysts exhibited white line intensities and edge energies close to that of the Pd foil (**Figure S6**), suggesting that the Pd atoms were well reduced to be metallic. The Pd K-edge XANES showed a slight shift to lower edge energies with increasing Cu wt% (**Table S2**). This shift suggests that Pd was most likely to be alloyed with Cu, as there would be electron transfer from Cu (electronegativity=1.9) to Pd (electronegativity=2.2), making Pd more electron rich upon alloy formation.[40]

The Fourier-transformed-EXAFS spectra further confirm the Pd−Cu alloy formation (**Figure 1c,d**).[41] When varying from Pd/Cu$_{(100:1)}$ to Pd/Cu$_{(1:100)}$, a continuous decrease of the Pd−Pd peak and increase of the Pd−Cu peak in Pd K-edge FT-EXAFS were observed (**Figure 1c**). At Pd/Cu$_{(1:100)}$, only Pd−Cu interaction existed, indicating the absence of Pd nanoclusters in the structure and the formation of Pd/Cu$_{(1:y)}$ SAA. In contrast, slight Cu−Cu interaction was observed in Pd/Cu$_{(100:1)}$ (**Figure 1d**), suggesting that small Cu nanoclusters/nanoparticles still existed. The quantitative Pd K edge FT-EXAFS fitting results (**Table S2, Figure S7**) showed that the Pd−Pd coordination number (CN) decreased from 10.6 of Pd/Cu$_{(100:1)}$ to 1.0 of Pd/Cu$_{(1:100)}$, whereas the Pd−Cu CN increased from 0.4 of Pd/Cu$_{(100:1)}$ to 11.0 of Pd/Cu$_{(1:100)}$). This confirms the gradual reduction of Pd cluster size inside the alloy with increasing Cu content. The fitted Pd−Pd radial distance in Pd/Cu$_{(1:100)}$ was 2.84 Å, which is larger than the typical Pd−Pd distance (~2.74 Å) in pure Pd metals or nanoparticles,[42] further suggesting the single atom state of Pd in the alloy. HAADF-STEM characterization provided visual evidence for the Pd single atoms, which were atomically dispersed on the Cu nanoparticle (**Figure 1e**). HAADF-STEM elemental mapping confirmed the distribution of Cu and Pd atoms across the Pd/Cu$_{(1:100)}$ surface (**Figure S8**).

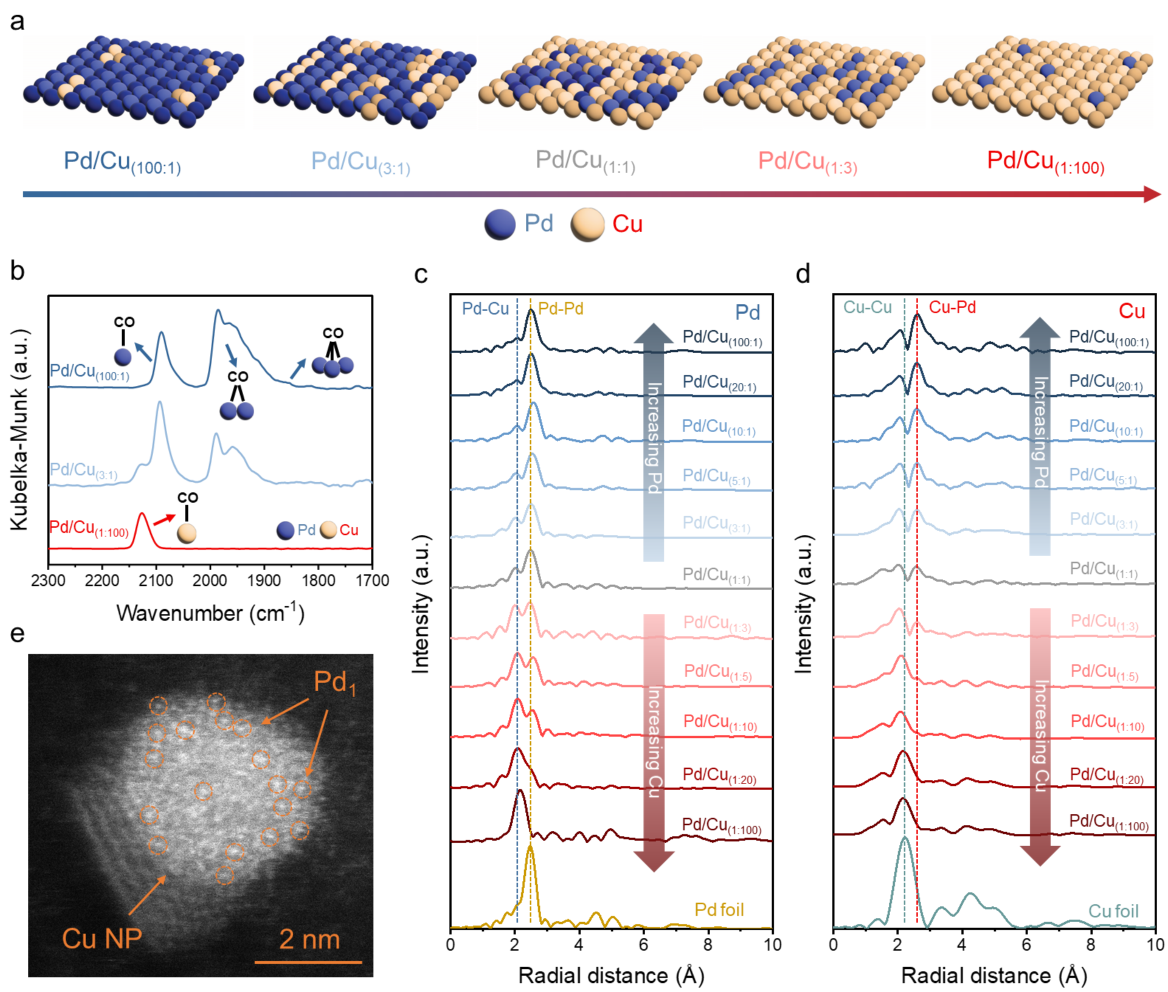


Figure 1. (a) Schematics of atomic structure for representative Pd/$Cu_{(x:y)}$ catalysts. (b) CO-DRIFTS of Pd/$Cu_{(x:y)}$ catalysts. (c) Pd and (d) Cu K-edge FT-EXAFS spectra of Pd/$Cu_{(x:y)}$ catalysts and foil references. (e) HAADF image of Pd/$Cu_{(1:100)}$ catalyst. Scale bar: 2 nm.

T-NRR performance of Pd/$Cu_{(x:y)}$ catalysts. Pd/$Cu_{(x:y)}$ catalysts with different Pd/Cu ratios exhibited varying T-NRR performances with Pd/$Cu_{(3:1)}$ showing the highest T-NRR efficiency (**Figure 2a,b**), almost completely reducing 50 mg/L NO_3^--N after 1 hour. Downsizing Pd from Pd/$Cu_{(100:1)}$ to Pd/$Cu_{(1:100)}$ first showed the increase in T-NRR performance until the ratio reached Pd/$Cu_{(3:1)}$. As the Cu content further increased, T-NRR performance continued to decrease. At the highest Cu content, the Pd/$Cu_{(1:100)}$ SAA only achieved 9% nitrate removal after 2 hours. This result indicates, within the Pd/Cu samples we tested, that Pd nanoparticles outperformed Pd single atoms in T-NRR.

The Pd/$Cu_{(3:1)}$ also generated the highest NO_2-N peak concentration (6.3 mg/L) within the 2-hour reaction (**Figure S9a,b**). From Pd/$Cu_{(100:1)}$ to Pd/$Cu_{(1:1)}$, the accumulation of NO_2-N was more significant with decreasing Pd wt%, which is ascribed to the fact that less Pd weakens the catalyst's capability to further reduce NO_2^-.[31, 43] Linear regression of Ln([NO_3-N]$_0$/[NO_3-N]) versus reaction time t (**Figure S9c,d**) provides the slope as the pseudo first-order rate constant k values (**Figure 2c**). A volcano-shape of k values was observed for Pd/$Cu_{(x:y)}$, with the highest k at Pd/$Cu_{(3:1)}$ (k = 0.056 min^{-1}). This optimal kinetic constant is comparable to previous Pd/Cu nitrate reduction catalysts on different supports (**Table S3**).[33, 44] The turnover frequency per mole of Pd (TOF_{Pd}) for different Pd/$Cu_{(x:y)}$ catalysts showed an optimal peak at Pd/$Cu_{(1:3)}$ (0.170 min^{-1}) (**Figure 2d**). The N concentrations (ppm) and N selectivity (%) at 2 hours shown in **Figure 2e,f** suggest that the Pd/$Cu_{(3:1)}$ achieved 77% N_2 selectivity and 19% NH_4^+ selectivity.

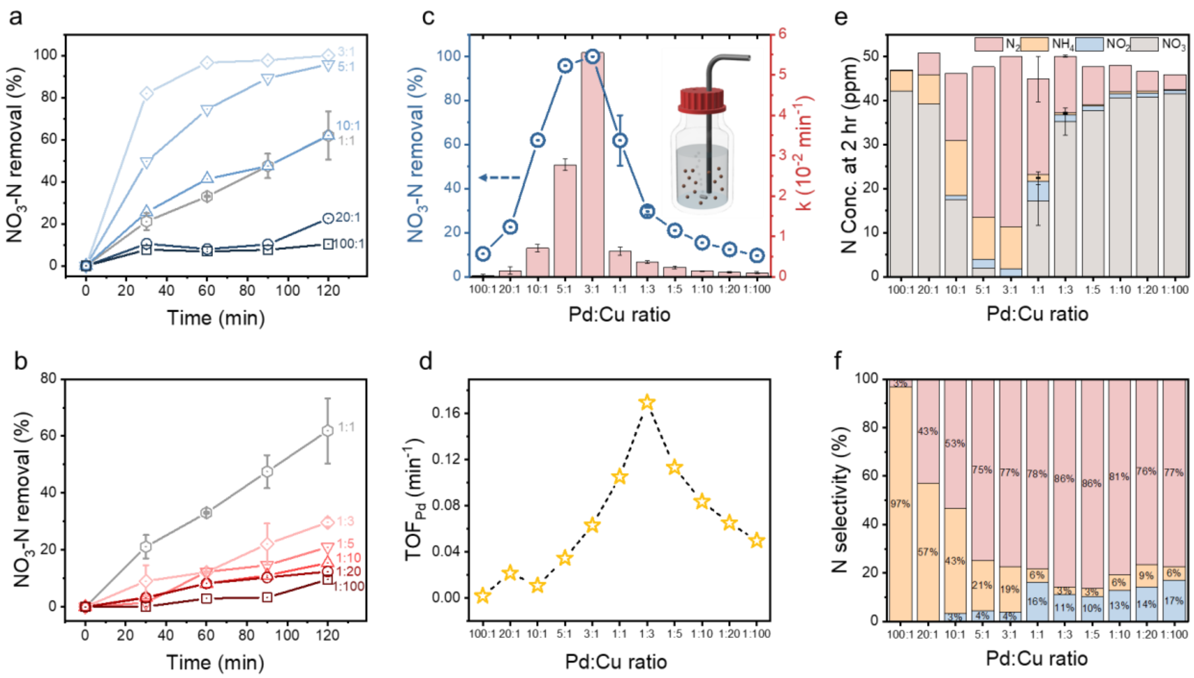


Figure 2. T-NRR nitrate-nitrogen (NO_3-N) removal rate (%) in the presence of Pd/$Cu_{(x:y)}$ from (a) Pd/$Cu_{(100:1)}$ to Pd/$Cu_{(1:1)}$ and (b) Pd/$Cu_{(1:1)}$ to Pd/$Cu_{(1:100)}$ at pH 6. (c) NO_3-N removal rate (%) in 2 hour (left y-axis) and pseudo first-order rate constant *k* values ($\times 10^{-2}$ min^{-1}, right y-axis) of different Pd/$Cu_{(x:y)}$ systems in T-NRR. *Inset*: schematic showing the batch system with *in situ* H_2 bubbling for T-NRR. (d) The turnover frequency with respect to per mole of Pd (TOF_{Pd}, min^{-1}) of Pd/$Cu_{(x:y)}$ catalysts. (e) N concentrations in 2 hour (ppm) of Pd/$Cu_{(x:y)}$ catalysts. Note that the initial NO_3^--N concentration (~50 mg/L) may vary slightly among systems, as the zero time point sample was taken right after adding the nitrate stock solution. (f) N selectivity (%) among reduced nitrogen species (N_2, NH_4^+, NO_2^-) of Pd/$Cu_{(x:y)}$ catalysts.

E-NRR performance of Pd/$Cu_{(x:y)}$ catalysts. We first performed linear sweep voltammetry (LSV) using Ar-saturated electrolytes with and without nitrate in a double compartment cell to evaluate the selectivity of Pd/$Cu_{(x:y)}$ towards E-NRR over HER (**Figure 3a-f**). The enhanced current density in the presence of nitrate suggests that nitrate was reduced on the catalyst loaded on the cathode. Note that a greater difference between the measured current density with and without nitrate at any given potential would indicate the higher performance of the catalyst toward E-NRR. In stark contrast to the T-NRR performance, under both low (50 mg/L NO_3^--N, or 3.57 mM, the same as in T-NRR) and high (7 g/L NO_3^--N, or 0.5 M) nitrate concentrations, Pd/$Cu_{(1:100)}$ SAA showed improved activity, and also exhibited the greatest activity over a wide range of potential windows as compared to other Pd/$Cu_{(x:y)}$ nanoalloy counterparts, including the additional control Pd/$Cu_{(0:100)}$ (with the same Cu contents as the Pd/$Cu_{(1:100)}$ but without any Pd). This finding suggests the unique role of Pd single atoms dispersed on Cu nanoparticles for E-NRR.

We further quantified nitrate removal rates and pseudo first-order rate constant *k* values for $Pd/Cu_{(x:y)}$ in E-NRR via chronoamperometry (CA) conducted at -0.4V *vs*. RHE for 2 hours with 50 mg/L NO_3^--N initial nitrate (**Figure 3g**). In contrast to the volcano-shape seen under T-NRR, a U-shape activity versus metal ratio relationship was obtained. The $Pd/Cu_{(1:100)}$ showed the highest nitrate removal rate of 41% within 2 hours which corresponded to the highest *k* of 0.0044 min^{-1}, in accordance with the trend of the selectivity performance obtained from LSV. The lowest nitrate removal rate (22.6%) and *k* (0.0021 min^{-1}) were obtained for $Pd/Cu_{(1:1)}$.

The TOF_{Pd} showed approximately 2 orders of magnitude increase from 0.04 min^{-1} for $Pd/Cu_{(100:1)}$ to 2 min^{-1} for $Pd/Cu_{(1:100)}$ (**Figure 3h**). The N_2 selectivity of $Pd/Cu_{(1:100)}$ reached 94%, which is the highest value among $Pd/Cu_{(x:y)}$ in both T-NRR (**Figure 2f**) and E-NRR (**Figure S10b**), and also higher than previously reported catalysts for E-NRR (**Table S4**). Among all catalysts tested, $Pd/Cu_{(1:100)}$ achieved the highest FE toward N_2 (**Table S5**), and were also the most effective in suppressing HER. It is noteworthy that, unlike T-NRR, E-NRR does not rely on Pd to supply protons through H_2 dissociation as the proton source is supplied through a water oxidation reaction. We postulate that this difference might contribute to the different performance of $Pd/Cu_{(1:100)}$ in T-NRR and E-NRR.

CA tests were additionally performed at a series of applied potentials for $Pd/Cu_{(1:100)}$ to evaluate the effects of potential on nitrate removal (**Figure 3i**). Both the nitrate removal rate and the rate constant followed a volcano-shape behavior with the greatest nitrate removal rate of 41% at 2 hours with a rate constant *k* of 0.0044 min^{-1} and N_2 selectivity of 94% (**Figure S12b**) at -0.4 V *vs.* RHE. The lower removal rates at potentials below -0.4 V are attributed to the lower amounts of charge available for the reaction as evidenced from current versus time curves measured at different potentials (**Figure S13**). In addition, the lower removal rates at potentials above -0.4 V are attributed to the dominance of HER over E-NRR (**Table S6**). Note that the final pH of the electrolytes increased from the initial 6 to ~13−13.5 after 2-hour reactions at different applied potentials (**Figure S14**) due to the consumption of protons by both E-NRR and HER.

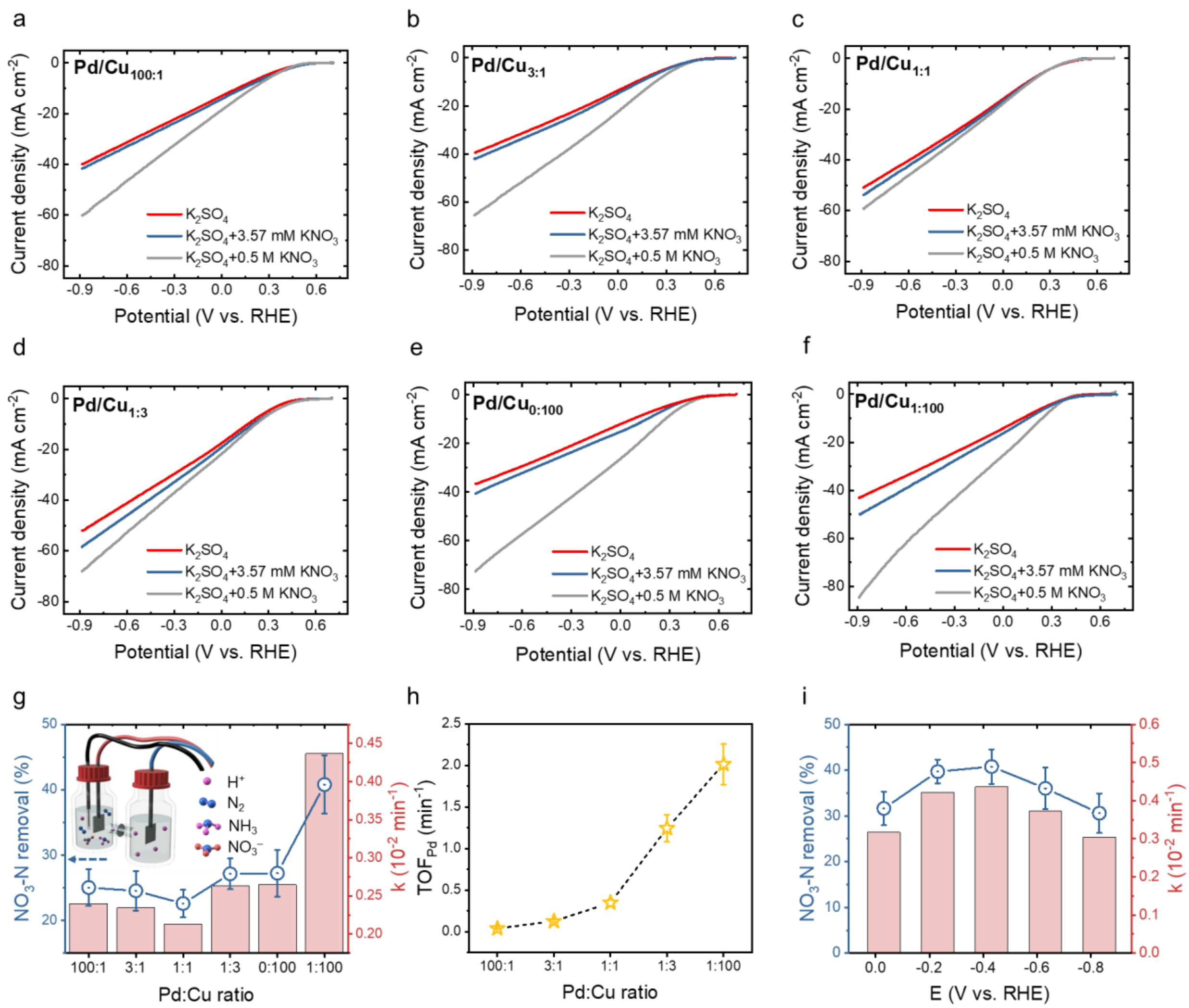


Figure 3. LSV curves of (a) Pd/Cu$_{(100:1)}$; (b) Pd/Cu$_{(3:1)}$; (c) Pd/Cu$_{(1:1)}$; (d) Pd/Cu$_{(1:3)}$; (e) Pd/Cu$_{(0:100)}$ control; and (f) Pd/Cu$_{(1:100)}$. (g) NO_3-N removal rate (%) at 2 hour (left y-axis) and pseudo first-order rate constant k values ($\times 10^{-2}$ min^{-1}, right y-axis) of different Pd/Cu$_{(x:y)}$ systems in E-NRR at -0.4V *vs*. RHE. *Inset*: two compartment electrochemical reaction cell schematic. (h) The TOF_{Pd} (min^{-1}) of Pd/Cu$_{(x:y)}$ catalysts. (i) NO_3-N removal rate (%) at 2 hour (left y-axis) and pseudo first-order rate constant k values ($\times 10^{-2}$ min^{-1}, right y-axis) of Pd/Cu$_{(1:100)}$ with varied potentials.

Mechanistic understanding. We performed DFT calculations to understand the reason behind the improved nitrate reduction activity and N_2 selectivity of Pd/Cu$_{(1:100)}$ SAA when switching from T-NRR to E-NRR. DFT predicts that NO_3^- adsorption is equally favorable on Pd_1 sites and on Cu NP sites of Pd/Cu$_{(1:100)}$. After adsorption to Pd_1 sites, NO_3^* (* denotes the adsorbed species) dissociates into NO_2^* then NO^* with activation barriers of 0.65 eV and 0.71 eV, respectively. Conversely, on the Cu NP sites, the activation barrier of NO_3^* dissociation is at least 1 eV higher than Pd_1 sites, which indicates that NRR occurs solely on Pd_1 sites. After screening multiple competitive pathways, including H-assisted and non-H assisted deoxygenation, we found that under E-NRR, H-assisted NO deoxygenation is preferred and occurs by hydrogenation of NO^* to HNO^* followed by subsequent cleavage of HNO^* to N^* and $OH^–$. After N^* forms, it preferentially migrates from the Pd_1 sites to the surrounding Cu NP matrix, where it either reacts with another N^* to form N_2 or with H to form NH^* and then follows the NH_3 pathway. The reaction map and main N reaction pathways for both T-NRR and E-NRR are shown in **Figure 4**.

T-NRR and E-NRR operate under different pH with distinct proton sources (*i.e.*, H_2O or H_3O^+), which affects the activity and selectivity of NRR (**Figure S15**). DFT calculations demonstrate that the higher $NO_3^–$ reduction activity of $Pd/Cu_{(1:100)}$ in E-NRR over T-NRR arises from higher NO_3^* stabilization (**Figure 4a**). Under E-NRR, the pH of the electrolyte rises rapidly to pH 13, which increases the free energy of the dissolved $NO_3^–$ by ~0.74 eV from the free energy at pH 6 of the T-NRR condition, and consequently stabilizes the adsorbed nitrate species. The increased stability of adsorbed NO_3^* and favorability of reaction under E-NRR condition means that NO_3^* dissociation (0.65 eV) is ~ 5.66×10^{12} times more likely than desorption (1.4 eV), and further surface reaction of the N species is favorable. Conversely, under T-NRR, NO_3^* desorption and dissociation are equally likely, with reaction energies of 0.66 eV and 0.65 eV, respectively. Thus, the adsorbed nitrate is just as possible to desorb into the solution as it is to dissociate under T-NRR conditions, hindering reaction progress.

DFT calculations also predict the higher N_2 selectivity of E-NRR *vs.* T-NRR for $Pd/Cu_{(1:100)}$ is because of lower $NO_2^–$ desorption rates and lower N_2 barriers as compared to the NH_3 path (**Figure 4b,c**). Under E-NRR conditions, $NO_2^–$ desorption and NO_2^* deoxygenation have roughly equivalent barriers (0.74 eV), which indicates equivalent reaction progression along each path. However, because the NO^* dissociation products are highly stabilized compared to the desorbed $NO_2^–$, and because the barriers are essentially equivalent, the dissociated product is the more likely result over time. The preference for NO^* arises because desorbed $NO_2^–$ could rapidly reabsorb without barrier and then decompose. Conversely, the acidic pH conditions of T-NRR stabilizes the desorption of $NO_2^–$ by 0.12 eV/pH, making $NO_2^–$ desorption slightly exothermic (-0.02 eV) at pH = 6, and likely than overcoming the 0.71 eV barrier for NO_2^* dissociation.

Moreover, the slightly acidic nature of the T-NRR pathway further reduces its selectivity to N_2 compared with alkaline E-NRR. Under both conditions, the reduced N^* hops preferentially to Cu sites (barrier of 0.02 eV) rather than remaining fixed on the Pd and undergoing hydrogenation (barriers of 0.98 eV for E-NRR and 0.83 eV for T-NRR); taken with the entropic preference for delocalization the N-N and N-H reactions are expected to occur on Cu sites (**Figure 4d**). On the Cu sites, the N^*-N^* recombination has a 0.29 eV barrier which is 0.49 eV lower than hydrogenation for E-NRR; thus DFT suggests a selective production of N_2 gas, in agreement with the experimental results. Moreover, the availability of H atoms from H_2 and H_3O^+, through decoupled electron/proton transfer, results in low activation barriers of hydrogenation steps under T-NRR by 0.29 eV (NH^* (0.60 eV), NH_2^* (0.74 eV), NH_3^* (0.85 eV) formation). Conversely, under E-NRR, only H_2O is available as the proton source, which provides fewer H atoms and increases the activation barrier of hydrogenation steps (NH^* (0.78 eV), NH_2^* (0.89 eV), NH_3^* (1.00 eV) formation). Overall, the higher pH, and H_2O as the only proton source contribute to the higher selectively towards N_2 of Pd/Cu$_{(1:100)}$ in E-NRR than in T-NRR.

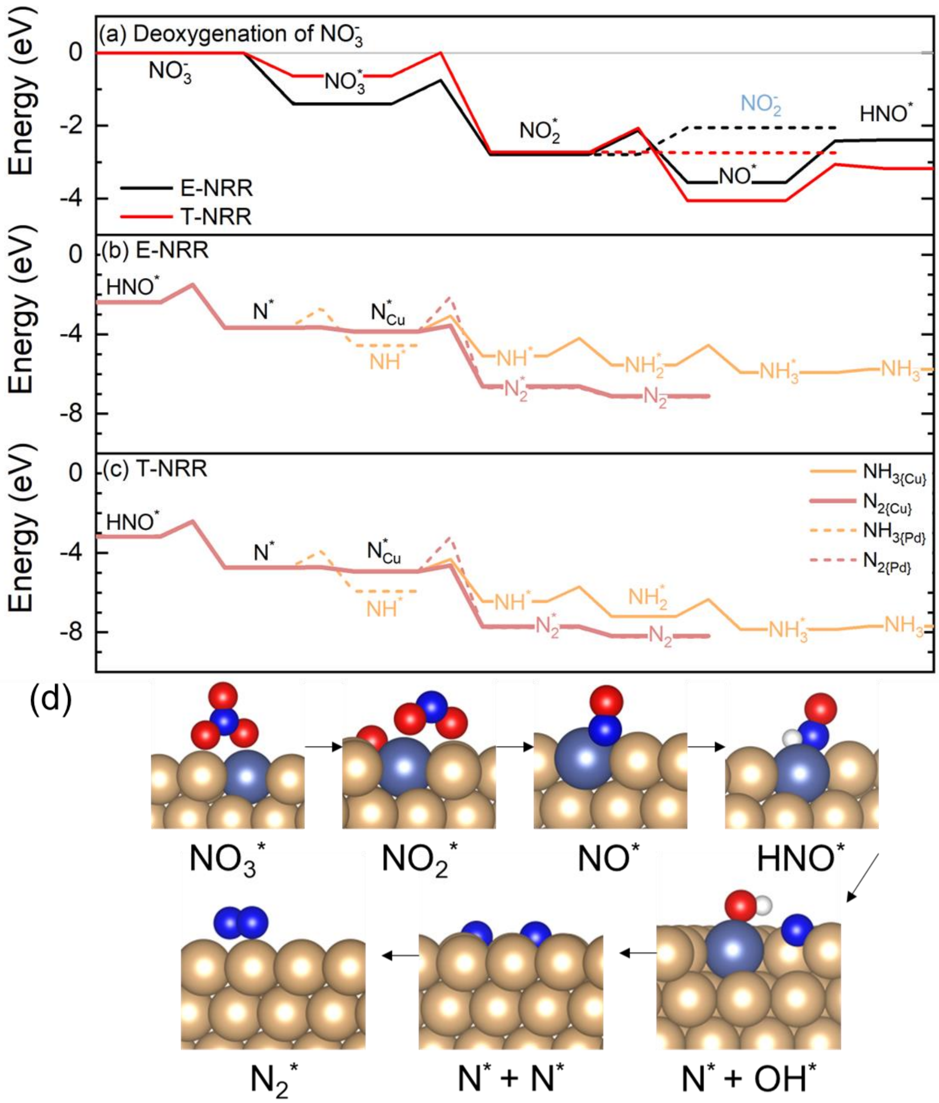


Figure 4. DFT calculations of (a) adsorption and deoxygenation of NO_3^- ($NO_3^- \rightarrow NO_3^*$, $NO_3^* \rightarrow NO_2^*$, $NO_2^* \rightarrow NO^*$, $NO^* \rightarrow HNO^*$) for T-NRR (red) and E-NRR (black) on the Pd_1 site of Pd/Cu$_{(1:100)}$ SAA. (b) HNO^* reduction on Pd_1 and Cu NP sites to N_2 and NH_3 for E-NRR at pH = 13 using H_2O as proton source. (c) HNO^* reduction on Pd_1 and Cu NP sites to N_2 and NH_3 for T-NRR at pH = 6 using H_3O^+ as proton source. (d) Meta-stable points along the minimum energy path for NO_3^- reduction to N_2. White, blue, red, golden, and blue-gray spheres correspond to H, N, O, Cu, and Pd atoms, respectively.

Conclusions

Using the Pd/Cu combination, we compared the T-NRR and E-NRR of $Pd/Cu_{(1:100)}$ SAA with $Pd/Cu_{(x:y)}$ nanoalloy counterparts. We found that $Pd/Cu_{(1:100)}$ SAA's activity increased significantly when switching from T-NRR to E-NRR. Under E-NRR, $Pd/Cu_{(1:100)}$ SAA exhibited the best activity and N_2 selectivity when compared to all nanoalloy counterparts. Further, DFT calculations indicate that for $Pd/Cu_{(1:100)}$, NRR occurs solely on the Pd_1 sites, from where the N^* species migrate to Cu NPs sites and combine into N_2. The improved performance and N_2 selectivity of $Pd/Cu_{(1:100)}$ in E-NRR over T-NRR arises from the higher stability of surface adsorbed NO_3^*, and a lower N_2 formation barrier than NH due to localized pH effects and ability to extract protons from water. Our work mechanistically compare the performance of $Pd/Cu_{(1:y)}$ SAA in thermo-hydrogenation *vs.* electro-reduction systems, particularly in nitrate reduction. Our findings are beneficial in obtaining a better understanding of the role of Pd single atoms over nanoparticle counterparts in hydrogenation processes, with demonstration of the effects of available energy sources (H_2 or electrical current) on reactions by altering localized pH and proton sources.

Supporting Information

The Supporting Information is available free of charge at XXX.

- Experimental methods, loading of Pd and Cu on $Pd/Cu_{(x:y)}$ catalysts, fitted XAS results, comparison of $Pd/Cu_{(3:1)}$ with reported Pd/Cu catalysts for T-NRR, comparison of $Pd/Cu_{(1:100)}$ with reported catalysts for E-NRR, schematic of the incipient wetness co-impregnation method, TEM, XRD, peak fittings for CO-DRIFTS spectra, XANES, EXAFS fitting, HAADF STEM elemental mapping, nitrite concentrations, E-NRR nitrogen distribution, current density changes, DFT calculations.

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References

(1) Hallberg, G. R. Nitrate in ground water in the United States. In *Developments in agricultural and managed forest ecology*, Vol. 21; Elsevier, 1989; pp 35-74.
(2) Puckett, L. J. *Nonpoint and point sources of nitrogen in major watersheds of the United States*; US Geological Survey, 1994; pp 1-9.
(3) Spalding, R. F.; Exner, M. E. Occurrence of nitrate in groundwater—a review. *J. Environ. Qual.* **1993**, *22* (3), 392-402.
(4) Ward, M. H.; Mark, S. D.; Cantor, K. P.; Weisenburger, D. D.; Correa-Villasenor, A.; Zahm, S. H. Drinking water nitrate and the risk of non-Hodgkin's lymphoma. *Epidemiology* **1996**, 465-471. Weyer, P. J.; Cerhan, J. R.; Kross, B. C.; Hallberg, G. R.; Kantamneni, J.; Breuer, G.; Jones, M. P.; Zheng, W.; Lynch, C. F. Municipal drinking water nitrate level and cancer risk in older women: the Iowa Women's Health Study. *Epidemiology* **2001**, 327-338.
(5) Smolders, A. J.; Lucassen, E. C.; Bobbink, R.; Roelofs, J. G.; Lamers, L. P. How nitrate leaching from agricultural lands provokes phosphate eutrophication in groundwater fed wetlands: the sulphur bridge. *Biogeochemistry* **2010**, *98* (1), 1-7.
(6) EPA. *Drinking Water Regulations and Health Advisories*; 1996.
(7) Heck, K. N.; Garcia-Segura, S.; Westerhoff, P.; Wong, M. S. Catalytic converters for water treatment. *Acc. Chem. Res.* **2019**, *52* (4), 906-915. Tokazhanov, G.; Ramazanova, E.; Hamid, S.; Bae, S.; Lee, W. Advances in the catalytic reduction of nitrate by metallic catalysts for high efficiency and N_2 selectivity: a review. *Chem. Eng. J.* **2020**, *384*, 123252.
(8) Jung, J.; Bae, S.; Lee, W. Nitrate reduction by maghemite supported Cu-Pd bimetallic catalyst. *Appl. Catal. B Environ.* **2012**, *127*, 148-158.
(9) Vorlop, K.-D.; Tacke, T. Erste schritte auf dem weg zur edelmetallkatalysierten nitrat-und nitrit-entfernung aus trinkwasser. *Chemieingenieurtechnik* **1989**, *61* (10), 836-837.
(10) Dima, G.; De Vooys, A.; Koper, M. Electrocatalytic reduction of nitrate at low concentration on coinage and transition-metal electrodes in acid solutions. *J. Electroanal. Chem.* **2003**, *554*, 15-23.
(11) Mácová, Z.; Bouzek, K.; Šerák, J. Electrocatalytic activity of copper alloys for NO_3^- reduction in a weakly alkaline solution. *Journal of applied electrochemistry* **2007**, *37* (5), 557-566.
(12) Reyter, D.; Belanger, D.; Roue, L. Elaboration of Cu−Pd films by coelectrodeposition: application to nitrate electroreduction. *J. Phys. Chem. C* **2009**, *113* (1), 290-297.
(13) Amertharaj, S.; Hasnat, M.; Mohamed, N. Electroreduction of nitrate ions at a platinum-copper electrode in an alkaline medium: Influence of sodium inositol phytate. *Electrochim. Acta* **2014**, *136*, 557-564.
(14) Mattarozzi, L.; Cattarin, S.; Comisso, N.; Gerbasi, R.; Guerriero, P.; Musiani, M.; Vazquez-Gomez, L.; Verlato, E. Electrodeposition of compact and porous Cu-Zn alloy electrodes and their use in the cathodic reduction of nitrate. *J. Electrochem. Soc.* **2015**, *162* (6), D236.
(15) Mattarozzi, L.; Cattarin, S.; Comisso, N.; Guerriero, P.; Musiani, M.; Vázquez-Gómez, L.; Verlato, E. Electrochemical reduction of nitrate and nitrite in alkaline media at CuNi alloy electrodes. *Electrochim. Acta* **2013**, *89*, 488-496.
(16) Da Cunha, M.; De Souza, J.; Nart, F. Reaction pathways for reduction of nitrate ions on platinum, rhodium, and platinum−rhodium alloy electrodes. *Langmuir* **2000**, *16* (2), 771-777.
(17) Garcia-Segura, S.; Lanzarini-Lopes, M.; Hristovski, K.; Westerhoff, P. Electrocatalytic reduction of nitrate: Fundamentals to full-scale water treatment applications. *Appl. Catal. B Environ.* **2018**, *236*, 546-568.
(18) Zhang, H.; Nguyen, H.; Vu, D.-A.; Bui, X.-N.; Pradhan, B. Forecasting monthly copper price: A comparative study of various machine learning-based methods. *Resour. Pol.* **2021**, *73*, 102189.
(19) Katsounaros, I. On the assessment of electrocatalysts for nitrate reduction. *Curr. Opin. Electrochem.* **2021**, *28*, 100721.
(20) Ghodbane, O.; Sarrazin, M.; Roué, L.; Bélanger, D. Electrochemical reduction of nitrate on pyrolytic graphite-supported Cu and Pd–Cu electrocatalysts. *J. Electrochem. Soc.* **2008**, *155* (6), F117. Su, J. F.; Ruzybayev, I.; Shah, I.; Huang, C. The electrochemical reduction of nitrate over micro-architectured metal electrodes with stainless steel scaffold. *Appl. Catal. B Environ.* **2016**, *180*, 199-209.
(21) Zhang, Z.; Xu, Y.; Shi, W.; Wang, W.; Zhang, R.; Bao, X.; Zhang, B.; Li, L.; Cui, F. Electrochemical-catalytic reduction of nitrate over Pd–Cu/γAl_2O_3 catalyst in cathode chamber: enhanced removal efficiency and N2 selectivity. *Chem. Eng. J.* **2016**, *290*, 201-208.
(22) Hannagan, R. T.; Giannakakis, G.; Flytzani-Stephanopoulos, M.; Sykes, E. C. H. Single-atom alloy catalysis. *Chem. Rev.* **2020**, *120* (21), 12044-12088.
(23) Boucher, M. B.; Zugic, B.; Cladaras, G.; Kammert, J.; Marcinkowski, M. D.; Lawton, T. J.; Sykes, E. C. H.; Flytzani-Stephanopoulos, M. Single atom alloy surface analogs in $Pd_{0.18}Cu_{15}$ nanoparticles for selective hydrogenation

reactions. *Phys. Chem. Chem. Phys.* **2013**, *15* (29), 12187-12196. Cao, X.; Fu, Q.; Luo, Y. Catalytic activity of Pd-doped Cu nanoparticles for hydrogenation as a single-atom-alloy catalyst. *Phys. Chem. Chem. Phys.* **2014**, *16* (18), 8367-8375. Kruppe, C. M.; Krooswyk, J. D.; Trenary, M. Selective hydrogenation of acetylene to ethylene in the presence of a carbonaceous surface layer on a Pd/Cu (111) single-atom alloy. *ACS Catal.* **2017**, *7* (12), 8042-8049. Kyriakou, G.; Boucher, M. B.; Jewell, A. D.; Lewis, E. A.; Lawton, T. J.; Baber, A. E.; Tierney, H. L.; Flytzani-Stephanopoulos, M.; Sykes, E. C. H. Isolated metal atom geometries as a strategy for selective heterogeneous hydrogenations. *Science* **2012**, *335* (6073), 1209-1212.

(24) Xing, F.; Jeon, J.; Toyao, T.; Shimizu, K.-i.; Furukawa, S. A Cu–Pd single-atom alloy catalyst for highly efficient NO reduction. *Chem. Sci.* **2019**, *10* (36), 8292-8298.

(25) Zhao, Z.; Lu, G. Cu-based single-atom catalysts boost electroreduction of CO_2 to CH_3OH: First-principles predictions. *J. Phys. Chem. C* **2019**, *123* (7), 4380-4387. Zheng, G.; Li, Y.; Qian, X.; Yao, G.; Tian, Z.; Zhang, X.; Chen, L. High-Throughput Screening of a Single-Atom Alloy for Electroreduction of Dinitrogen to Ammonia. *ACS Appl. Mater. Inter.* **2021**, *13* (14), 16336-16344.

(26) Chen, T.; Li, H.; Ma, H.; Koper, M. T. Surface modification of Pt (100) for electrocatalytic nitrate reduction to dinitrogen in alkaline solution. *Langmuir* **2015**, *31* (10), 3277-3281.

(27) Calle-Vallejo, F.; Huang, M.; Henry, J. B.; Koper, M. T.; Bandarenka, A. S. Theoretical design and experimental implementation of Ag/Au electrodes for the electrochemical reduction of nitrate. *Phys. Chem. Chem. Phys.* **2013**, *15* (9), 3196-3202.

(28) Yang, J.; Calle‐Vallejo, F.; Duca, M.; Koper, M. T. Electrocatalytic Reduction of Nitrate on a Pt Electrode Modified by p‐Block Metal Adatoms in Acid Solution. *ChemCatChem* **2013**, *5* (7), 1773-1783.

(29) Chen, F.-Y.; Wu, Z.-Y.; Gupta, S.; Rivera, D. J.; Lambeets, S. V.; Pecaut, S.; Kim, J. Y. T.; Zhu, P.; Finfrock, Y. Z.; Meira, D. M. Efficient conversion of low-concentration nitrate sources into ammonia on a Ru-dispersed Cu nanowire electrocatalyst. *Nat. Nanotech.* **2022**, 1-9.

(30) Hörold, S.; Vorlop, K.-D.; Tacke, T.; Sell, M. Development of catalysts for a selective nitrate and nitrite removal from drinking water. *Catal. Today* **1993**, *17* (1-2), 21-30.

(31) Shin, H.; Jung, S.; Bae, S.; Lee, W.; Kim, H. Nitrite reduction mechanism on a Pd surface. *Environ. Sci. Technol.* **2014**, *48* (21), 12768-12774.

(32) Barrabés, N.; Just, J.; Dafinov, A.; Medina, F.; Fierro, J.; Sueiras, J.; Salagre, P.; Cesteros, Y. Catalytic reduction of nitrate on Pt-Cu and Pd-Cu on active carbon using continuous reactor: The effect of copper nanoparticles. *Appl. Catal. B Environ.* **2006**, *62* (1-2), 77-85.

(33) Trawczyński, J.; Gheek, P.; Okal, J.; Zawadzki, M.; Gomez, M. I. Reduction of nitrate on active carbon supported Pd-Cu catalysts. *Appl. Catal. A General* **2011**, *409*, 39-47.

(34) Cao, Y.; Chen, B.; Guerrero-Sánchez, J.; Lee, I.; Zhou, X.; Takeuchi, N.; Zaera, F. Controlling selectivity in unsaturated aldehyde hydrogenation using single-site alloy catalysts. *ACS Catal.* **2019**, *9* (10), 9150-9157. Kim, J.; Roh, C. W.; Sahoo, S. K.; Yang, S.; Bae, J.; Han, J. W.; Lee, H. Highly durable platinum single‐atom alloy catalyst for electrochemical reactions. *Adv. Energy Mater.* **2018**, *8* (1), 1701476. Pei, G. X.; Liu, X. Y.; Yang, X.; Zhang, L.; Wang, A.; Li, L.; Wang, H.; Wang, X.; Zhang, T. Performance of Cu-alloyed Pd single-atom catalyst for semihydrogenation of acetylene under simulated front-end conditions. *ACS Catal.* **2017**, *7* (2), 1491-1500. Sun, G.; Zhao, Z.-J.; Mu, R.; Zha, S.; Li, L.; Chen, S.; Zang, K.; Luo, J.; Li, Z.; Purdy, S. C. Breaking the scaling relationship via thermally stable Pt/Cu single atom alloys for catalytic dehydrogenation. *Nat. Comm.* **2018**, *9* (1), 1-9.

(35) Anastasopoulos, A.; Hannah, L.; Hayden, B. E. High throughput optimisation of PdCu alloy electrocatalysts for the reduction of nitrate ions. *J. Catal.* **2013**, *305*, 27-35.

(36) Petro, J.; Mallat, T.; Szabo, S.; Hange, F. Anodic dissolution of palladium+copper alloys. *J. Electroanal. Chem. Interf. Electrochem.* **1984**, *160* (1-2), 289-297.

(37) Bertarione, S.; Scarano, D.; Zecchina, A.; Johanek, V.; Hoffmann, J.; Schauermann, S.; Frank, M. M.; Libuda, J.; Rupprechter, G.; Freund, H.-J. Surface reactivity of Pd nanoparticles supported on polycrystalline substrates as compared to thin film model catalysts: infrared study of CO adsorption. *J. Phys. Chem. B* **2004**, *108* (11), 3603-3613.

(38) Kuhn, W. K.; Szanyi, J.; Goodman, D. W. CO adsorption on Pd (111): the effects of temperature and pressure. *Surf. Sci.* **1992**, *274* (3), L611-L618.

(39) Dulaurent, O.; Courtois, X.; Perrichon, V.; Bianchi, D. Heats of adsorption of CO on a Cu/Al_2O_3 catalyst using FTIR spectroscopy at high temperatures and under adsorption equilibrium conditions. *J. Phys. Chem. B* **2000**, *104* (25), 6001-6011.

(40) Cho, E.-J.; Lee, S.; Oh, S.-J.; Han, M.; Lee, Y.; Whang, C. Unoccupied states and charge transfer in Cu-Pd alloys studied by bremsstrahlung isochromat spectroscopy, X-ray photoelectron spectroscopy, and L III absorption spectroscopy. *Phys. Rev. B* **1995**, *52* (23), 16443.
(41) Castegnaro, M. V.; Gorgeski, A.; Alves, M. d. C. M.; Morais, J. Carbon Supported Pd-Cu Nanoalloys: Support and Valence Band Structure Influence on the Reduction and Oxidation Reactions. *Nanoscale Adv.* **2021**. Yang, G.; Kuwahara, Y.; Mori, K.; Louis, C.; Yamashita, H. Pd–Cu Alloy Nanoparticles Confined within Mesoporous Hollow Carbon Spheres for the Hydrogenation of CO_2 to Formate. *J. Phys. Chem. C* **2021**, *125* (7), 3961-3971.
(42) Bugaev, A. L.; Usoltsev, O. A.; Guda, A. A.; Lomachenko, K. A.; Pankin, I. A.; Rusalev, Y. V.; Emerich, H.; Groppo, E.; Pellegrini, R.; Soldatov, A. V. Palladium carbide and hydride formation in the bulk and at the surface of palladium nanoparticles. *J. Phys. Chem. C* **2018**, *122* (22), 12029-12037. Shimizu, K.-i.; Kubo, T.; Satsuma, A.; Kamachi, T.; Yoshizawa, K. Surface oxygen atom as a cooperative ligand in Pd nanoparticle catalysis for selective hydration of nitriles to amides in water: experimental and theoretical studies. *ACS Catal.* **2012**, *2* (12), 2467-2474.
(43) Jung, S.; Bae, S.; Lee, W. Development of Pd–Cu/hematite catalyst for selective nitrate reduction. *Environ. Sci. Technol.* **2014**, *48* (16), 9651-9658.
(44) Chen, Y.-X.; Zhang, Y.; Chen, G.-H. Appropriate conditions or maximizing catalytic reduction efficiency of nitrate into nitrogen gas in groundwater. *Water Res.* **2003**, *37* (10), 2489-2495. Soares, O. S. G.; Órfão, J. J.; Pereira, M. F. Nitrate reduction in water catalysed by Pd–Cu on different supports. *Desalination* **2011**, *279* (1-3), 367-374.